

# **DESIGN AND ANALYSIS OF ALGORITHM (CS-327)**

## **LECTURE – 4**

### **STEP FREQUENCY COUNT TABLE :-**

Step frequency count method is one of the method to analyze the time complexity of an algorithm. In this method we count the number of times each instructions is executed. On the basis of such count we calculate the time complexity.

1. Design an algorithm to find the largest element of an array of size (N=10,N=100, N).  
Analyze it through SFC table and draw analysis table at end.

### **Solution of Problem-1:-**

#### **Formal description of algorithm includes:-**

1. Name of algorithm.
2. An array of integers.
3. Pre condition (state ) means state of machine prior to execution of algorithm(A[1].....A[10]).
4. State after execution of algorithm(Max/Largest element of an array).
5. Body of algorithm.

**Step Frequency Count Table (SFC) for N=10**

| Algorithm                 | Step(S) | Frequency (F) |     | Count (C = S X F) |     |
|---------------------------|---------|---------------|-----|-------------------|-----|
|                           |         | Min           | Max | Min               | Max |
| Start algorithm Max       |         |               |     |                   |     |
| Max $\leftarrow$ A(1)     | 2       | 1             | 1   | 2                 | 2   |
| For i $\leftarrow$ 2 to 9 |         |               |     |                   |     |
| if A(i) > Max then        | 2       | 9             | 9   | 18                | 18  |
| Max $\leftarrow$ A(i)     | 2       | 0             | 9   | 0                 | 18  |
| End if                    |         |               |     |                   |     |
| End for                   |         |               |     |                   |     |
| Out Max                   | 1       | 1             | 1   | 1                 | 1   |
| End algorithm             |         |               |     |                   |     |
| Total steps               |         |               |     | 21                | 39  |

**Step Frequency Count Table (SFC) for N=100**

| Algorithm                  | Step(S) | Frequency (F) |     | Count (C = S X F) |     |
|----------------------------|---------|---------------|-----|-------------------|-----|
|                            |         | Min           | Max | Min               | Max |
| Start algorithm Max        |         |               |     |                   |     |
| Max $\leftarrow$ A(1)      | 2       | 1             | 1   | 2                 | 2   |
| For i $\leftarrow$ 2 to 99 |         |               |     |                   |     |
| if A(i) > Max then         | 2       | 99            | 99  | 198               | 198 |
| Max $\leftarrow$ A(i)      | 2       | 0             | 99  | 0                 | 198 |
| End if                     |         |               |     |                   |     |
| End for                    |         |               |     |                   |     |
| Out Max                    | 1       | 1             | 1   | 1                 | 1   |
| End algorithm              |         |               |     |                   |     |
| Total steps                |         |               |     | 201               | 399 |

### Step Frequency Count Table (SFC) for N

| Algorithm                 | Step(S) | Frequency (F) |     | Count (C = S X F) |        |
|---------------------------|---------|---------------|-----|-------------------|--------|
|                           |         | Min           | Max | Min               | Max    |
| Start algorithm Max       |         |               |     |                   |        |
| Max $\leftarrow$ A(1)     | 2       | 1             | 1   | 2                 | 2      |
| For i $\leftarrow$ 2 to N |         |               |     |                   |        |
| if A(i) > Max then        | 2       | N-1           | N-1 | 2(N-1)            | 2(N-1) |
| Max $\leftarrow$ A(i)     | 2       | 0             | N-1 | 0                 | 2(N-1) |
| End if                    |         |               |     |                   |        |
| End for                   |         |               |     |                   |        |
| Out Max                   | 1       | 1             | 1   | 1                 | 1      |
| End algorithm             |         |               |     |                   |        |
| Total steps               |         |               |     | 2N+1              | 4N-1   |

### Analysis Table of (N=10,N=100,N)

| No. of steps | Min Run time     | Max Runtime |
|--------------|------------------|-------------|
| RTC(10)      | 21               | 39          |
| RTC(100)     | 201              | 399         |
| RTC(N)       | 2N+1             | 4N-1        |
| ARTC         | W(N) or $\pi(N)$ | O(N)        |

2. Design an algorithm to find the largest or smallest element of an array of size (N=10,N=100, N).Analyze it through SFC table and draw analysis table at end.

**Step Frequency Count Table (SFC) for N=10**

| Algorithm           | Step(S) | Frequency (F) |     | Count (C = S X F) |     |
|---------------------|---------|---------------|-----|-------------------|-----|
|                     |         | Min           | Max | Min               | Max |
| Start algorithm Max |         |               |     |                   |     |
| Max ← A(1)          | 2       | 1             | 1   | 2                 | 2   |
| Min ← A(1)          | 2       | 1             | 1   | 2                 | 2   |
| For i ← 2 to 9      |         |               |     |                   |     |
| if A(i) > Max then  | 2       | 9             | 9   | 18                | 18  |
| Max ← A(i)          | 2       | 0             | 9   | 0                 | 18  |
| End if              |         |               |     |                   |     |
| if A(i) < Min then  | 2       | 9             | 9   | 18                | 18  |
| Min ← A(i)          | 2       | 0             | 9   | 0                 | 18  |
| Endif               |         |               |     |                   |     |
| End for             |         |               |     |                   |     |
| Out Max             | 1       | 1             | 1   | 1                 | 1   |
| Out Min             | 1       | 1             | 1   | 1                 | 1   |
| End algorithm       |         |               |     |                   |     |
| Total steps         |         |               |     | 21                | 39  |

**Note :-**

In the above SFC table we count either min/max steps because one condition will execute at a time.

**Step Frequency Count Table (SFC) for N=100**

| Algorithm                  | Step(S) | Frequency (F) |     | Count (C = S X F) |     |
|----------------------------|---------|---------------|-----|-------------------|-----|
|                            |         | Min           | Max | Min               | Max |
| Start algorithm Max        |         |               |     |                   |     |
| Max $\leftarrow$ A(1)      | 2       | 1             | 1   | 2                 | 2   |
| Min $\leftarrow$ A(1)      | 2       | 1             | 1   | 2                 | 2   |
| For i $\leftarrow$ 2 to 99 |         |               |     |                   |     |
| if A(i) > Max then         | 2       | 99            | 99  | 198               | 198 |
| Max $\leftarrow$ A(i)      | 2       | 0             | 99  | 0                 | 198 |
| End if                     |         |               |     |                   |     |
| if A(i) < Min then         | 2       | 99            | 99  | 198               | 198 |
| Min $\leftarrow$ A(i)      | 2       | 0             | 99  | 0                 | 198 |
| Endif                      |         |               |     |                   |     |
| End for                    |         |               |     |                   |     |
| Out Max                    | 1       | 1             | 1   | 1                 | 1   |
| Out Min                    | 1       | 1             | 1   | 1                 | 1   |
| End algorithm              |         |               |     |                   |     |
| Total steps                |         |               |     | 201               | 399 |

**Note :-**

In the above SFC table we count either min/max steps because one condition will execute at a time.

**Step Frequency Count Table (SFC) for N**

| Algorithm                 | Step(S) | Frequency (F) |     | Count (C = S X F) |        |
|---------------------------|---------|---------------|-----|-------------------|--------|
|                           |         | Min           | Max | Min               | Max    |
| Start algorithm Max       |         |               |     |                   |        |
| Max $\leftarrow$ A(1)     | 2       | 1             | 1   | 2                 | 2      |
| Min $\leftarrow$ A(1)     | 2       | 1             | 1   | 2                 | 2      |
| For i $\leftarrow$ 2 to N |         |               |     |                   |        |
| if A(i) > Max then        | 2       | N-1           | N-1 | 2(N-1)            | 2(N-1) |
| Max $\leftarrow$ A(i)     | 2       | 0             | N-1 | 0                 | 2(N-1) |
| End if                    |         |               |     |                   |        |
| if A(i) < Min then        | 2       | N-1           | N-1 | 2(N-1)            | 2(N-1) |
| Min $\leftarrow$ A(i)     | 2       | 0             | N-1 | 0                 | 2(N-1) |
| Endif                     |         |               |     |                   |        |
| End for                   |         |               |     |                   |        |
| Out Max                   | 1       | 1             | 1   | 1                 | 1      |
| Out Min                   | 1       | 1             | 1   | 1                 | 1      |
| End algorithm             |         |               |     |                   |        |
| Total steps               |         |               |     | 2N+1              | 4N-1   |

**Note :-**

In the above SFC table we count either min/max steps because one condition will execute at a time.

**Analysis Table of (N=10,N=100,N)**

| No. of steps | Min Run time       | Max Runtime |
|--------------|--------------------|-------------|
| RTC(10)      | 21                 | 39          |
| RTC(100)     | 201                | 399         |
| RTC(N)       | $2N+1$             | $4N-1$      |
| ARTC         | $W(N)$ or $\pi(N)$ | $O(N)$      |

3. Design an algorithm to find the Fibonacci series up to (N=10) and analyze it through SFC table, also show working steps.

**Solution of problem3:-****Step Frequency Count Table (SFC)**

| Algorithm                        | Step(S) | Frequency (F) |     | Count (C = S X F) |           |
|----------------------------------|---------|---------------|-----|-------------------|-----------|
|                                  |         | Min           | Max | Min               | Max       |
| Start algorithm Fibonacci series |         |               |     |                   |           |
| a ← 0                            | 1       | 1             | 1   | 1                 | 1         |
| b ← 1                            | 1       | 1             | 1   | 1                 | 1         |
| Sum ← 0                          | 1       | 1             | 1   | 1                 | 1         |
| for i 1 ← 10                     |         |               |     |                   |           |
| Sum ← a+b                        | 3       | 10            | 10  | 30                | 30        |
| a ← b                            | 2       | 10            | 10  | 20                | 20        |
| b ← sum                          | 2       | 10            | 10  | 20                | 20        |
| Out Fibonacci series             | 10      | 1             | 1   | 10                | 10        |
| End for                          |         |               |     |                   |           |
| End algorithm                    |         |               |     |                   |           |
| <b>Total steps</b>               |         |               |     | <b>83</b>         | <b>83</b> |

**Working Steps :-**

| a  | b  | Sum |
|----|----|-----|
| 0  | 1  | 1   |
| 1  | 1  | 2   |
| 1  | 2  | 3   |
| 2  | 3  | 5   |
| 3  | 5  | 8   |
| 5  | 8  | 13  |
| 8  | 13 | 21  |
| 13 | 21 | 34  |
| 21 | 34 | 55  |
| 34 | 55 | 89  |

4. Read the following algorithm (pseudo code) carefully.

Algo JustDoIT (X,N)

K  $\leftarrow$  0

For i  $\leftarrow$  1 to N do

K  $\leftarrow$  K + X[ i ]

A[ i ]  $\leftarrow$  K / i

End for i

Return array A

End Algo Just Do It

Using SFC-Table, determine the counts of the total number of primitive steps performed during execution of algorithm, in best case and worst case. And then determine how many primitive steps would be performed if the value of N is 10; and what will be returned by algorithm if the array X has the elements : 5,7,3,5,5,5,12,6,6 and 16. Also show the working steps in table.

#### Solution of Problem-4:-

#### Step Frequency Count Table (SFC)

| Algorithm                    | Step(S) | Frequency (F) |     | Count (C = S X F) |           |
|------------------------------|---------|---------------|-----|-------------------|-----------|
|                              |         | Min           | Max | Min               | Max       |
| Algo Just DoIT (X,N)         |         |               |     |                   |           |
| K $\leftarrow$ 0             | 1       | 1             | 1   | 1                 | 1         |
| For i $\leftarrow$ 1 to N do |         |               |     |                   |           |
| K $\leftarrow$ K + X[ i ]    | 3       | 10            | 10  | 30                | 30        |
| A[ i ] $\leftarrow$ K / i    | 3       | 10            | 10  | 30                | 30        |
| End for i                    |         |               |     |                   |           |
| Return array A               | 10      | 1             | 1   | 10                | 10        |
| End Algo Just Do It          |         |               |     |                   |           |
| <b>Total steps</b>           |         |               |     | <b>71</b>         | <b>71</b> |

#### Working Steps :-

| X[ ] | i  | k  | A[ ] |
|------|----|----|------|
| 5    | 1  | 5  | 5    |
| 7    | 2  | 12 | 6    |
| 3    | 3  | 15 | 5    |
| 5    | 4  | 20 | 5    |
| 5    | 5  | 25 | 5    |
| 5    | 6  | 30 | 5    |
| 12   | 7  | 42 | 6    |
| 6    | 8  | 48 | 6    |
| 6    | 9  | 54 | 6    |
| 16   | 10 | 70 | 7    |



5. Design an algorithm to find the table of a no# up to (N=10) and analyze it through SFC table, also show working steps.

**Solution of problem3:-**

**Step Frequency Count Table (SFC)**

| Algorithm             | Step(S) | Frequency (F) |     | Count (C = S X F) |     |
|-----------------------|---------|---------------|-----|-------------------|-----|
|                       |         | Min           | Max | Min               | Max |
| Start algorithm Table |         |               |     |                   |     |
| i ← 1                 | 1       | 1             | 1   | 1                 | 1   |
| j ← 1                 | 1       | 1             | 1   | 1                 | 1   |
| table ← 0             | 1       | 1             | 1   | 1                 | 1   |
| for i 1 ← 10          |         |               |     |                   |     |
| for j 1 ← 10          |         |               |     |                   |     |
| tab ← i * j           | 3       | 100           | 100 | 300               | 300 |
| Out tab               | 100     | 1             | 1   | 100               | 100 |
| End for               |         |               |     |                   |     |
| End algorithm         |         |               |     |                   |     |
| Total steps           |         |               |     | 403               | 403 |

**Working Steps :-**

| i  | j  |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|----|-----|
| 1  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10  |
| 2  | 2  | 4  | 6  | 8  | 10 | 12 | 14 | 16 | 18 | 20  |
| 3  | 3  | 6  | 9  | 12 | 15 | 18 | 21 | 24 | 27 | 30  |
| 4  | 4  | 8  | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40  |
| 5  | 5  | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50  |
| 6  | 6  | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60  |
| 7  | 7  | 14 | 21 | 28 | 32 | 36 | 40 | 44 | 48 | 52  |
| 8  | 8  | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80  |
| 9  | 9  | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90  |
| 10 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |